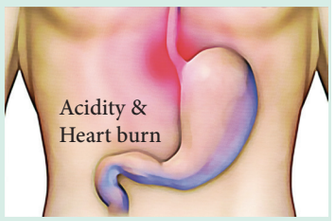
Let us calculate the amount of acid neutralised by an antacid that contains 250 mg of aluminium hydroxide and 250 mg of magnesium hydroxide.

Active Compound	Mass in (mg)	Molecular mass ($g\ mol^{-1}$)	No. of moles of active compound	No. of moles OH^-
$Al(OH)_3$	250	78	0.0032	0.0096
$Mg(OH)_2$	250	58	0.0043	0.0086
Total no. of moles of OH^- ion from one tablet				0.0182

One tablet of above composition will neutralise 0.0182 mole of HCl for a person with gastric acid content of 0.1 mole. One tablet can be used to neutralize the excess acid which will bring the concentration back to normal level. (0.1 – 0.018 = 0.082 M)



1.4.1 Avogadro Number:

The total number of entities present in one mole of any substance is equal to 6.022×10^{23}. This number is called Avogadro number which is named after the Italian physicist Amedeo Avogadro who proposed that equal volume of all gases under the same conditions of temperature and pressure contain equal number of molecules. Avogadro number does not have any unit.



Lorenzo Romano Amedeo Carlo Avogadro (1776-1856)

He is known for the Avogadro's hypothesis. In honour of his contributions, the number of fundamental particles in a mole of substance was named as Avogadro number. Though Avogadro didn't predict the number of particles in equal volumes of gas, his hypothesis did lead to the eventual determination of the number as 6.022×10^{23} Rudolf Clausius, with his kinetic theory of gases, provided evidence for Avogadro's law.

In a chemical reaction, atoms or molecules react in a specific ratio. Let us consider the following examples

Reaction 1 : $C + O_2 \rightarrow CO_2$

Reaction 2 : $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$

In the first reaction, one carbon atom reacts with one oxygen molecule to give one carbon dioxide molecule. In the second reaction, one molecule of methane burns with two molecules of oxygen to give one molecule of carbon dioxide and two molecules of water. It is clear that the ratio of reactants is based on the number of molecules. Even though the ratio is based on the number of molecules it is practically difficult to count the number of molecules. Because of this reason it is beneficial to use **'mole'** concept rather than the actual number of molecules to quantify the reactants and the products. We can explain the first reaction as one mole of carbon reacts with one mole of oxygen to give one mole of carbon dioxide and the second reaction as one mole of methane burns with two moles of oxygen to give one mole of carbon dioxide and two moles of water. When only atoms are involved, scientists also use the term one gram atom instead of one mole.

11th Std Chemistry 001-037.indd 7 23-12-2021 15:03:19